Litu Ou

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EDUCATION

University of Edinburgh

- PhD in Computer Science, Sep 2025 – Expected Sep 2029
  Advisor: Prof. Mirella Lapata
- BSc (Hons) in Artificial Intelligence and Computer Science, First-Class Honours, Sep 2019 – May 2024

RESEARCH SUMMARY

My research focuses on enabling language models to effectively organize and reason over long contexts. My interests span hierarchical representations, retrieval and information synthesis, agentic search, and post-training and test-time control. My current work investigates how models can learn to incrementally build persistent structures over long documents and adaptively traverse them to support downstream reasoning.

PUBLICATIONS

- **Storyline Trees: Hierarchical Representations for Long-Form Narratives** (*Findings of EMNLP 2026*)
  Litu Ou, Mirella Lapata
- **BrowseConf: Confidence-Guided Test-Time Scaling for Web Agents** (*Findings of ACL 2026*)
  Litu Ou*, Kuan Li*, Huifeng Yin*, Liwen Zhang, Zhongwang Zhang, Xixi Wu, et al.
- **Context-Aware Hierarchical Merging for Long Document Summarization** (*Findings of ACL 2025*)
  Litu Ou, Mirella Lapata
- **Revisiting Text Ranking in Deep Research** (*SIGIR 2026*)
  Chuan Meng, **Litu Ou**, Sean MacAvaney, Jeff Dalton
- **WebSailor: Navigating Super-human Reasoning for Web Agent** (*arXiv preprint)*
  Kuan Li*, Zhongwang Zhang*, Huifeng Yin*, Liwen Zhang*, **Litu Ou***, Jialong Wu, et al.
- **Tongyi DeepResearch Technical Report (***arXiv preprint***)**
  Tongyi DeepResearch Team
- **WebSailor-V2: Bridging the Chasm to Proprietary Agents via Synthetic Data and Scalable Reinforcement Learning** (*ICLR 2026*)
  Kuan Li, Zhongwang Zhang, Huifeng Yin, Rui Ye, Yida Zhao, Liwen Zhang, **Litu Ou**, Dingchu Zhang, et al.
- **Uncertainty-Aware Gradient Signal-to-Noise Data Selection for Instruction Tuning** (*EMNLP 2026 Main Conference*)
  Zhihang Yuan, Cheng Yue, Long Huang, **Litu Ou**, Lei Shi
- **Specializing Smaller Language Models towards Multi-Step Reasoning** (*ICML 2023, Oral*)
  Yao Fu, Hao Peng, **Litu Ou**, Ashish Sabharwal, Tushar Khot
- **Chain-of-Thought Hub: A Continuous Effort to Measure Large Language Models' Reasoning Performance** (*ICML 2023*)
  Yao Fu, **Litu Ou,** Yuhao Wan, Mingyu Chen, Hao Peng, Tushar Khot

- **ReSum: Unlocking Long-Horizon Search Intelligence via Context Summarization** (*arXiv preprint*)
  Xixi Wu, Kuan Li, Yida Zhao, Liwen Zhang, **Litu Ou**, et al.
- **Measuring Context Utilization on Recent Open-Source Long Context LLMs (***ICLR Blog Posts 2025*)
  Litu Ou
- **ToLeaP: Rethinking Development of Tool Learning with Large Language Models** (*arXiv preprint*)
  Haotian Chen*, Zijun Song*, Boye Niu*, Zhang Ke*, **Litu Ou***, Yaxi Lu, Zhong Zhang, et al.

* Equal contribution.

CURRENT RESEARCH

Long-Context Representation, Memory, and Reasoning

- Investigating how language models incrementally construct, update, and organize persistent hierarchical representations as new information becomes available.
- Developing adaptive retrieval mechanisms to identify and integrate evidence distributed across long contexts for downstream reasoning.
- Exploring post-training approaches for jointly learning memory construction and retrieval through task-level feedback.

RESEARCH EXPERIENCE

Research Intern, Tongyi Lab, Alibaba Group, Jun 2025 – Sep 2025

- Core contributor to Tongyi DeepResearch and its technical report, focusing on post-training and test-time improvements for LLM-based web agents in long-horizon search and open-ended task-solving settings.
- Led research on confidence-aware evaluation and test-time control for LLM search agents, resulting in BrowseConf (Findings of ACL 2026).
- Contributed to RL synthetic data generation and training stability for long-horizon web agents, monitoring tool-call success rates, timeouts, and long rollouts, and investigating termination and reward strategies.

Remote Research Assistant, Prof. Zhiyuan Liu's lab, Tsinghua University, Aug 2024 – May 2025

- Conducted research on building a universal benchmark and evaluation pipeline for assessing tool-using capabilities in large language models.
- Contributed to infrastructure engineering for scalable synthetic data generation and agentic reinforcement learning.

Junior Research Assistant, ILCC, University of Edinburgh, Jun 2023 – Dec 2024
Advisor: Prof. Mirella Lapata

- Designed retrieval-enhanced summarization pipelines for book-length documents, leveraging QA-based blueprint generation and content selection to improve factual consistency and faithfulness for long document summarization.
- Proposed a context-aware hierarchical merging strategy to improve faithfulness in summarizing book-length documents. This work resulted in a Findings of ACL 2025 publication.

Software Engineer Intern, Applied Machine Learning Department, ByteDance Inc., Sep 2021 – Jul 2022

- Contributed to the development of a creative advertising platform using React and TypeScript, and supported the development of an internal machine learning platform.

AWARDS

- **Young Software Engineer of the Year Awards 2024**, Finalist
  Sole nominee from the University of Edinburgh, based on undergraduate thesis work.

ACADEMIC SERVICE

- **Conference Reviewer:** NeurIPS 2026 Evaluations and Datasets Track
- **Workshop Reviewer:** LCFM @ NeurIPS 2026; LASS @ CIKM 2026; LLA @ COLM 2026; LLA @ ICLR 2026

TALKS & TUTORIALS

- **Agentic Search: Foundations, Optimization, and Long-Horizons** (CIKM 2026 Tutorial, Rome, Italy)
  Co-presenter with Chuan Meng, Sean MacAvaney, and Jeff Dalton; presented long-horizon agentic search and deep-research agents. Tutorial Website

TEACHING EXPERIENCE

- **Teaching Assistant**, *Foundations of Natural Language Processing*, Jan – Jun 2026
- **Demonstrator**, *Foundations of Natural Language Processing* (2024); I*ntroduction to Algorithms and Data Structures* (2023)
- **Tutor**, *Introduction to Computation* (2020)

SKILLS

- **Programming:** Python, C++, Java, JavaScript, TypeScript
- **ML Frameworks:** PyTorch, TensorFlow, Hugging Face, vLLM, SGLang
- **Research Engineering:** large-scale experimentation, evaluation pipelines, ablation studies, long-context and agentic LLM evaluation
- **Systems:** Linux, remote GPU servers, reproducible research workflows

HOBBIES

Table tennis, piano, swimming, hiking, movies